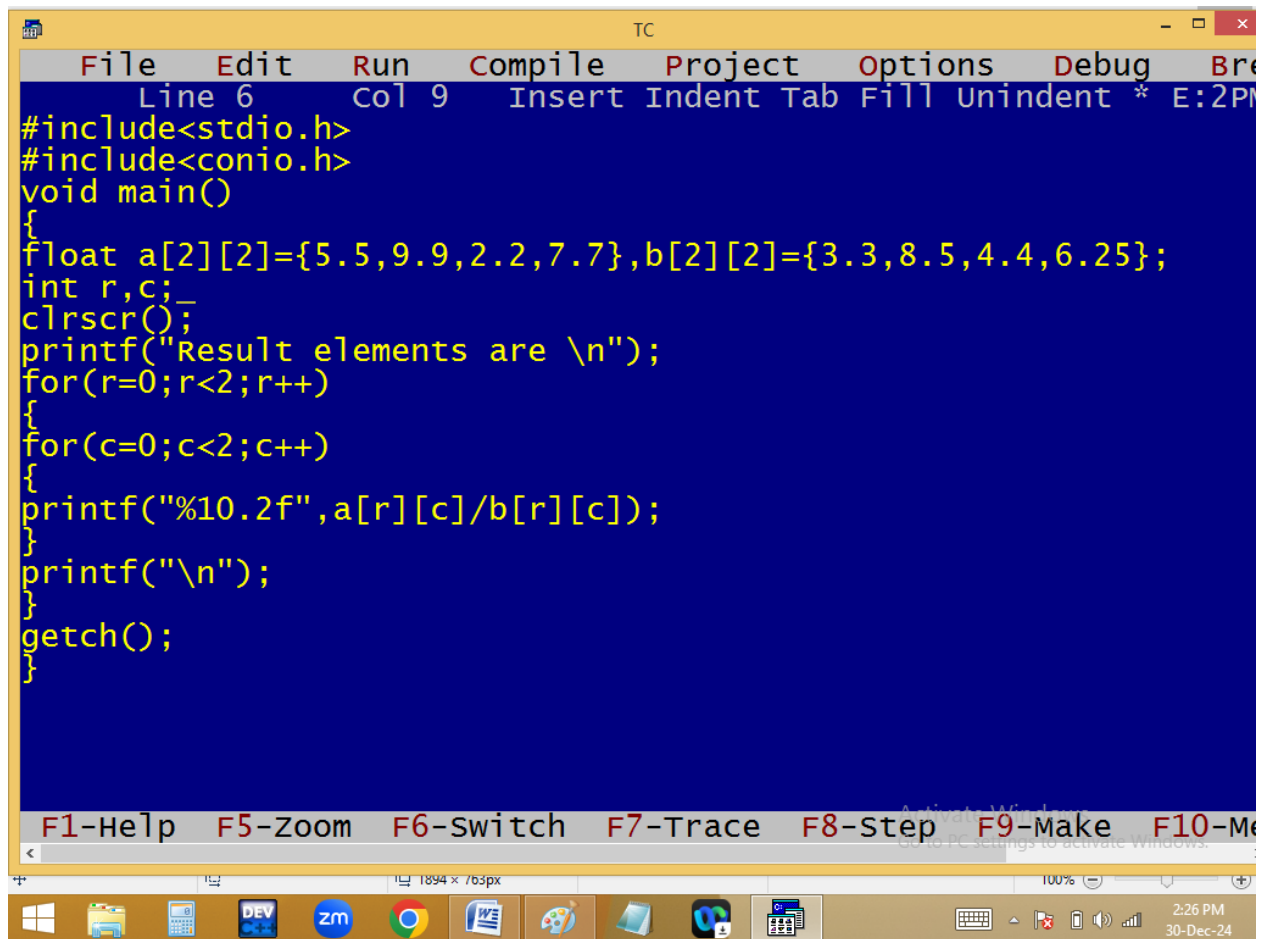


Finding fraction of $n \times n$ matrix:

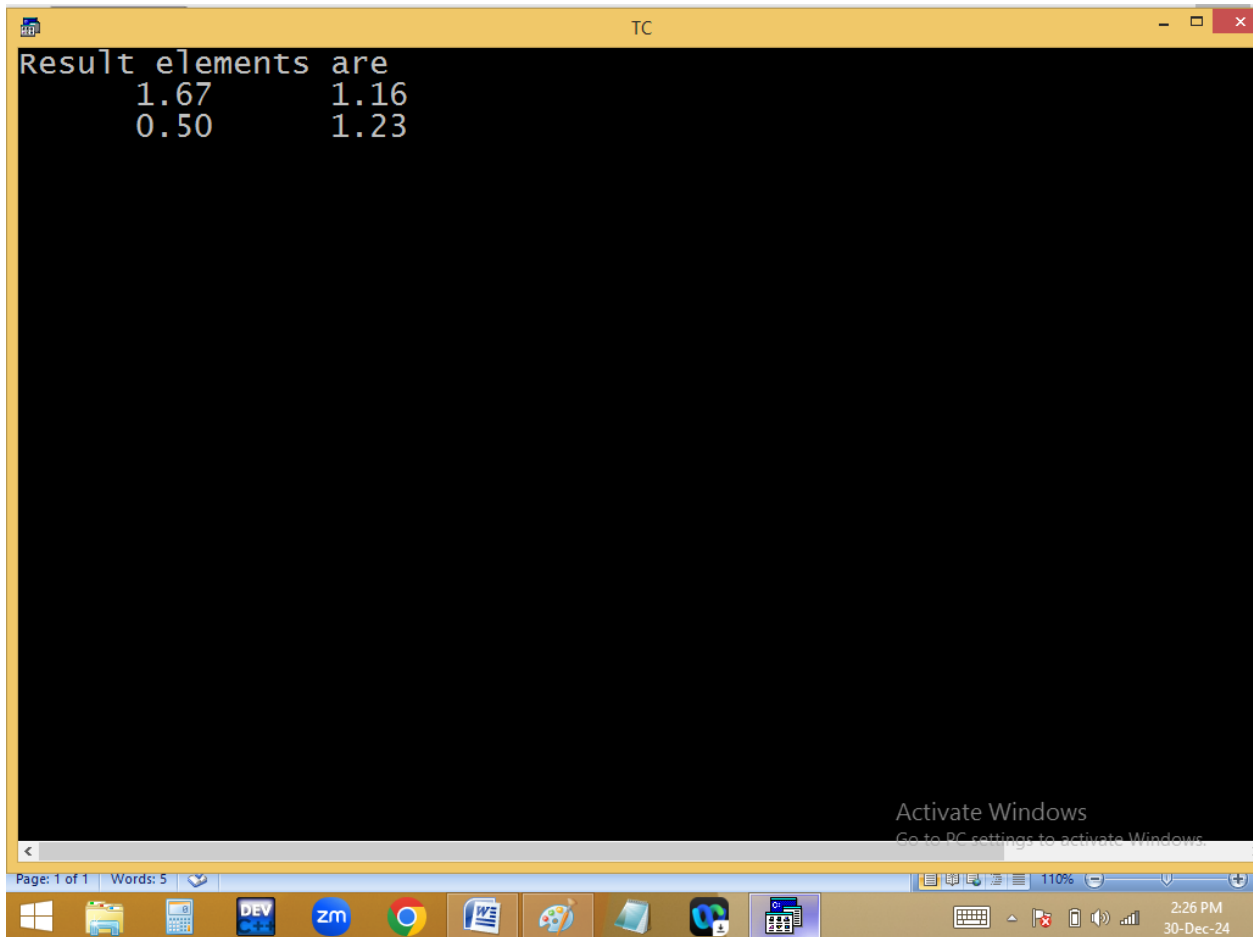


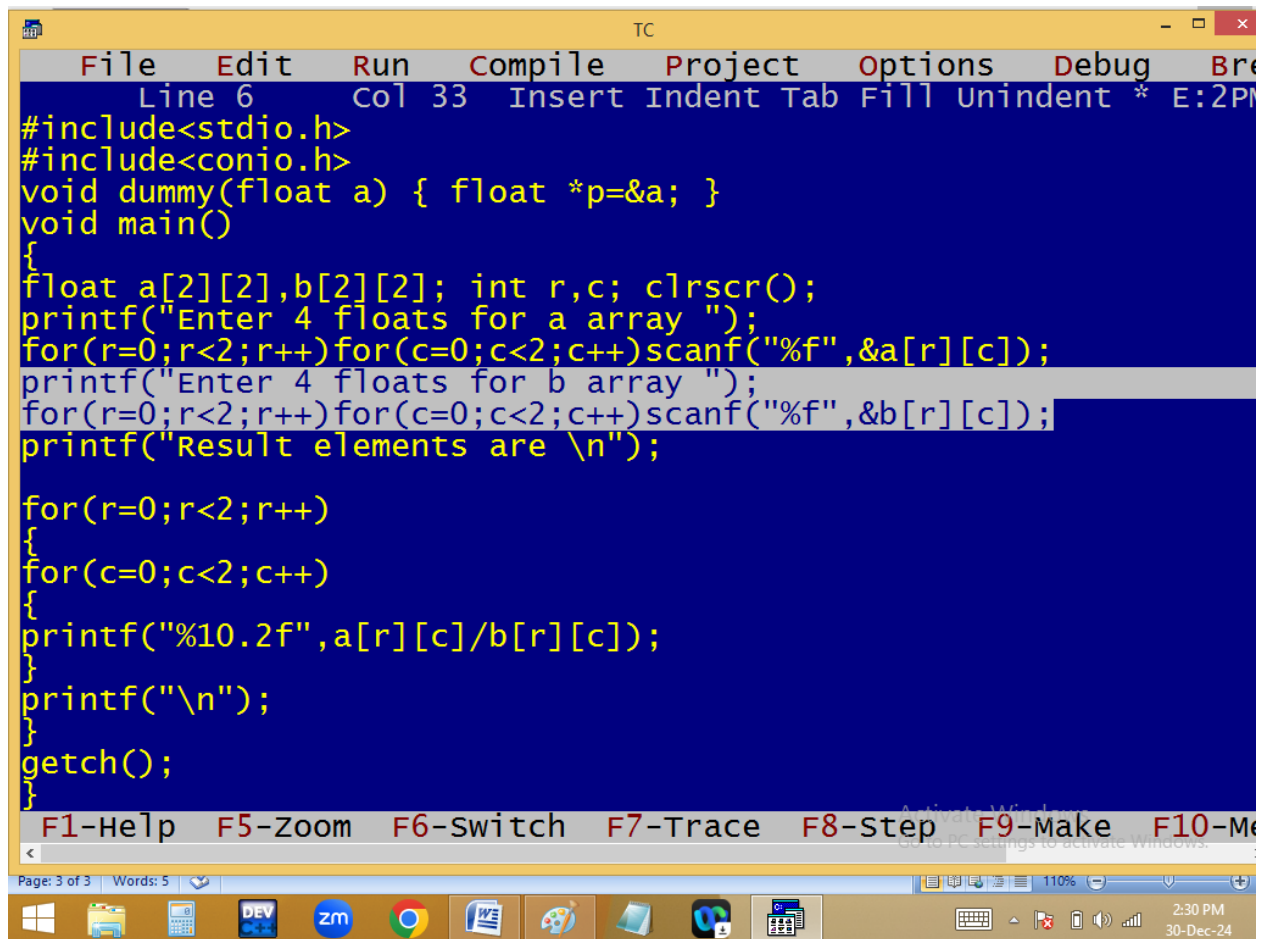
```
TC
File Edit Run Compile Project Options Debug Bro
Line 6 Col 9 Insert Indent Tab Fill Unindent * E:2P
#include<stdio.h>
#include<conio.h>
void main()
{
float a[2][2]={5.5,9.9,2.2,7.7},b[2][2]={3.3,8.5,4.4,6.25};
int r,c;
clrscr();
printf("Result elements are \n");
for(r=0;r<2;r++)
{
for(c=0;c<2;c++)
{
printf("%10.2f",a[r][c]/b[r][c]);
printf("\n");
}
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

1894 x 763px 100%

2:26 PM 30-Dec-24





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break". The status bar at the top indicates "Line 6", "Col 33", and "Insert Indent Tab Fill Unindent * E:2P". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, defines a `dummy` function, and a `main` function. The `main` function declares two 2x2 float arrays, `a` and `b`, and uses nested loops to input values for each. It then prints the result of dividing the elements of array `a` by the corresponding elements of array `b` using `printf` with a format of `%10.2f`. The program ends with `getch()`. The bottom status bar shows "Page: 3 of 3" and "Words: 5". The Windows taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, Paint, and a folder icon. The system clock shows "2:30 PM" and "30-Dec-24".

```
File Edit Run Compile Project Options Debug Break
Line 6 Col 33 Insert Indent Tab Fill Unindent * E:2P
#include<stdio.h>
#include<conio.h>
void dummy(float a) { float *p=&a; }
void main()
{
float a[2][2],b[2][2]; int r,c; clrscr();
printf("Enter 4 floats for a array ");
for(r=0;r<2;r++)for(c=0;c<2;c++)scanf("%f",&a[r][c]);
printf("Enter 4 floats for b array ");
for(r=0;r<2;r++)for(c=0;c<2;c++)scanf("%f",&b[r][c]);
printf("Result elements are \n");

for(r=0;r<2;r++)
{
for(c=0;c<2;c++)
{
printf("%10.2f",a[r][c]/b[r][c]);
}
printf("\n");
}
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Page: 3 of 3 Words: 5
2:30 PM
30-Dec-24
```

```
TC
Enter 4 floats for a array 1 2 3 4
Enter 4 floats for b array 5 6 7 8
Result elements are
    0.20    0.33
    0.43    0.50
```

Activate Windows
Go to PC settings to activate Windows.

Page: 4 of 4 Words: 5 110% 2:30 PM 30-Dec-24

```
for(r=0;r<2;r++)
{
for(c=0;c<2;c++)
{
p(a[r][c]/b[r][c]);
}
p("\n");
}
```

a		b			
5.5	9.9	3.3	8.5	1.66	1.16
0,0	0,1	0,0	0,1		
2.2	7.7	4.4	6.25	0.5	1.23
1,0	1,1	1,0	1,1		

Matrix multiplication:

a		b	
1 0,0	2 0,1	5 0,0	6 0,1
3 1,0	4 1,1	7 1,0	8 1,1

$$1*5+2*7=19$$

$$1*6+2*8=22$$

$$3*5+4*7=43$$

$$3*6+4*8=50$$

```

TC
File Edit Run Compile Project Options Debug Bro
Line 17 Col 2 Insert Indent Tab Fill Unindent * E:2P
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],b[10][10]; int nr,nc,r,c,k,s; clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d int's for a array ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
printf("Enter %d int's for b array ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&b[r][c]);
printf("Result elements are \n");

for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{for(k=s=0;k<nc;k++){s+=a[r][k]*b[k][c];}
printf("%4d",s);
}
printf("\n");
}
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

```

```
TC
Enter no of rows and columns 2 2
Enter 4 int's for a array 1 2 3 4
Enter 4 int's for b array 5 6 7 8
Result elements are
  19  22
  43  50
```

Activate Windows
Go to PC settings to activate Windows.

Page: 6 of 7 Words: 7

2:51 PM
30-Dec-24

```

TC
Enter no of rows and columns 2 3
Enter 6 int's for a array 1 2 3 4 5 6
Enter 6 int's for b array 1 2 3 4 5 6
Result elements are
23561-68491755
-18408-136893522

```

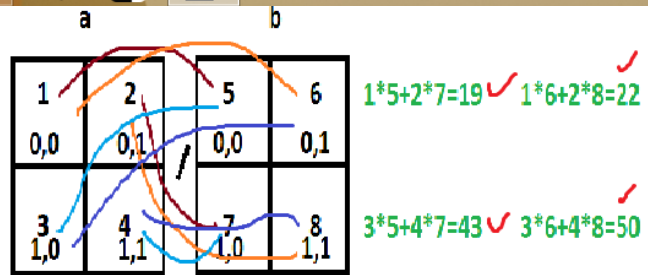
Page: 8 of 8 Words: 7

2:53 PM 30-Dec-24

```

for(r=0;r<2;r++)
{
for(c=0;c<2;c++)
{
for(s=k=0;k<2;k++)
{
s+=a[r][k]*b[k][c];
}
p(s);
}
p("\n");
}

```



r	c	k	s
0	0	012	0+1*5=5+2*7=19 ✓
0	1	012	0+1*6=6+2*8=22 ✓
1	0	012	0+3*5=15+4*7=43 ✓
1	1	012	0+3*6=18+4*8=50 ✓

3-dimensional arrays:

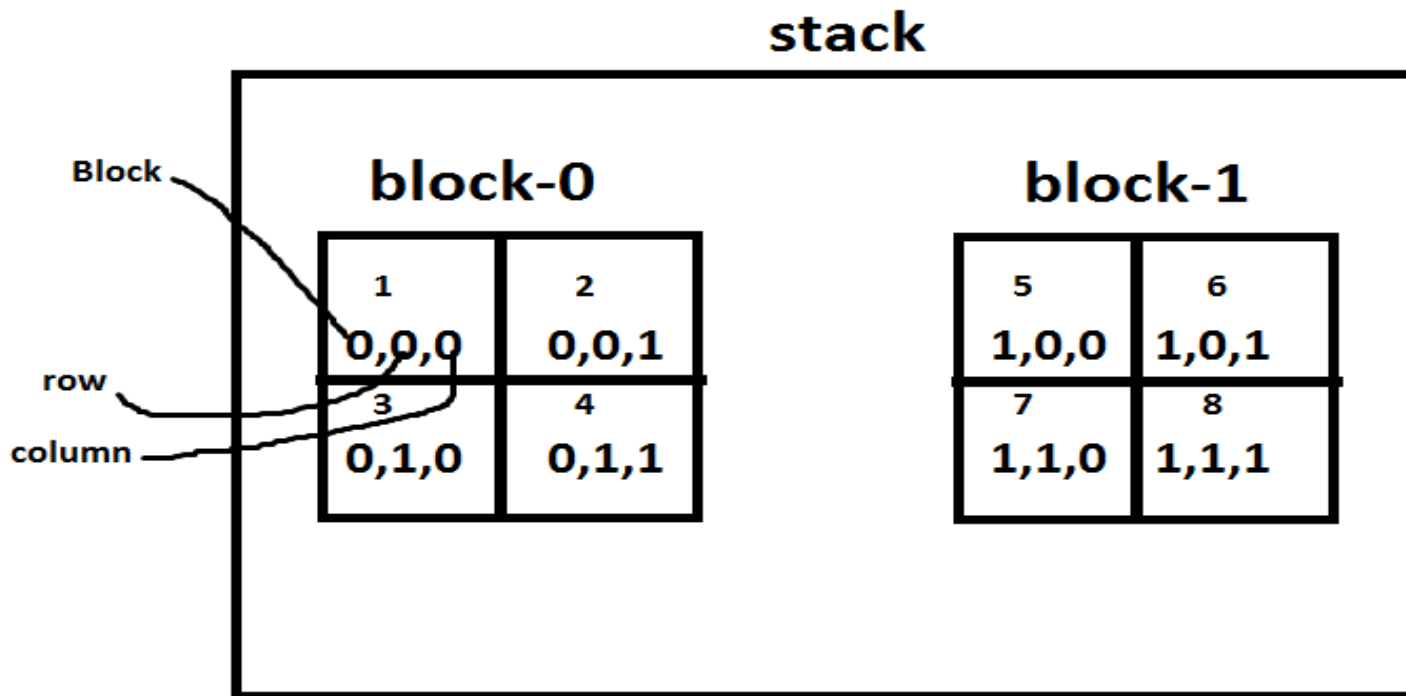
An array with several blocks, rows and columns.

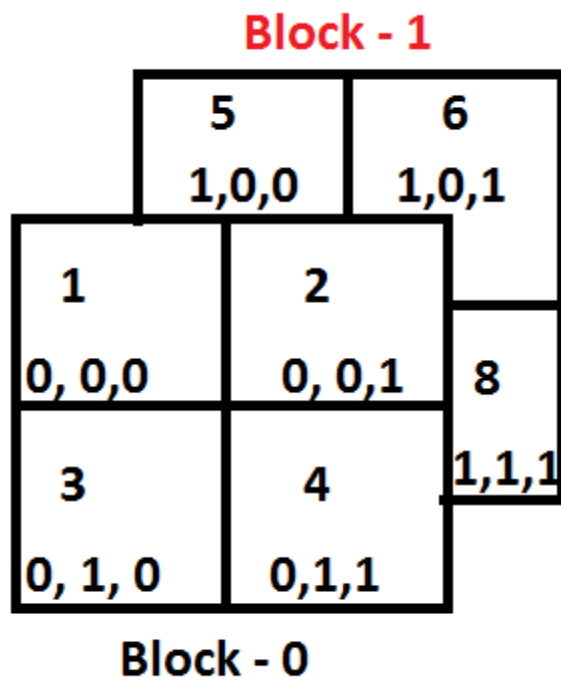
An array with 3 subscripting operators **[] [] []**.

Syntax:

datatype variable [blocks] [rows] [columns];

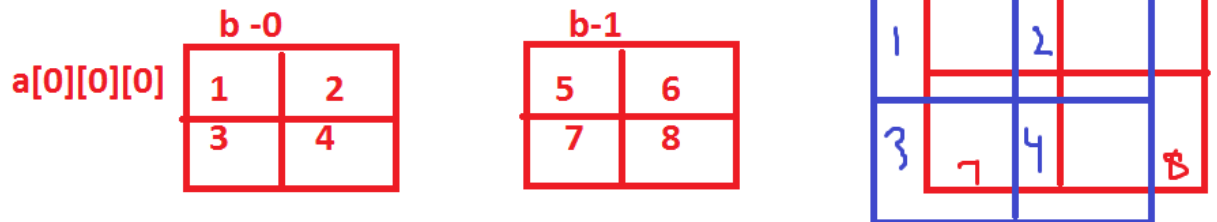
Eg: int a[2][2][2]={1,2,3,4,5,6,7,8};





Eq:

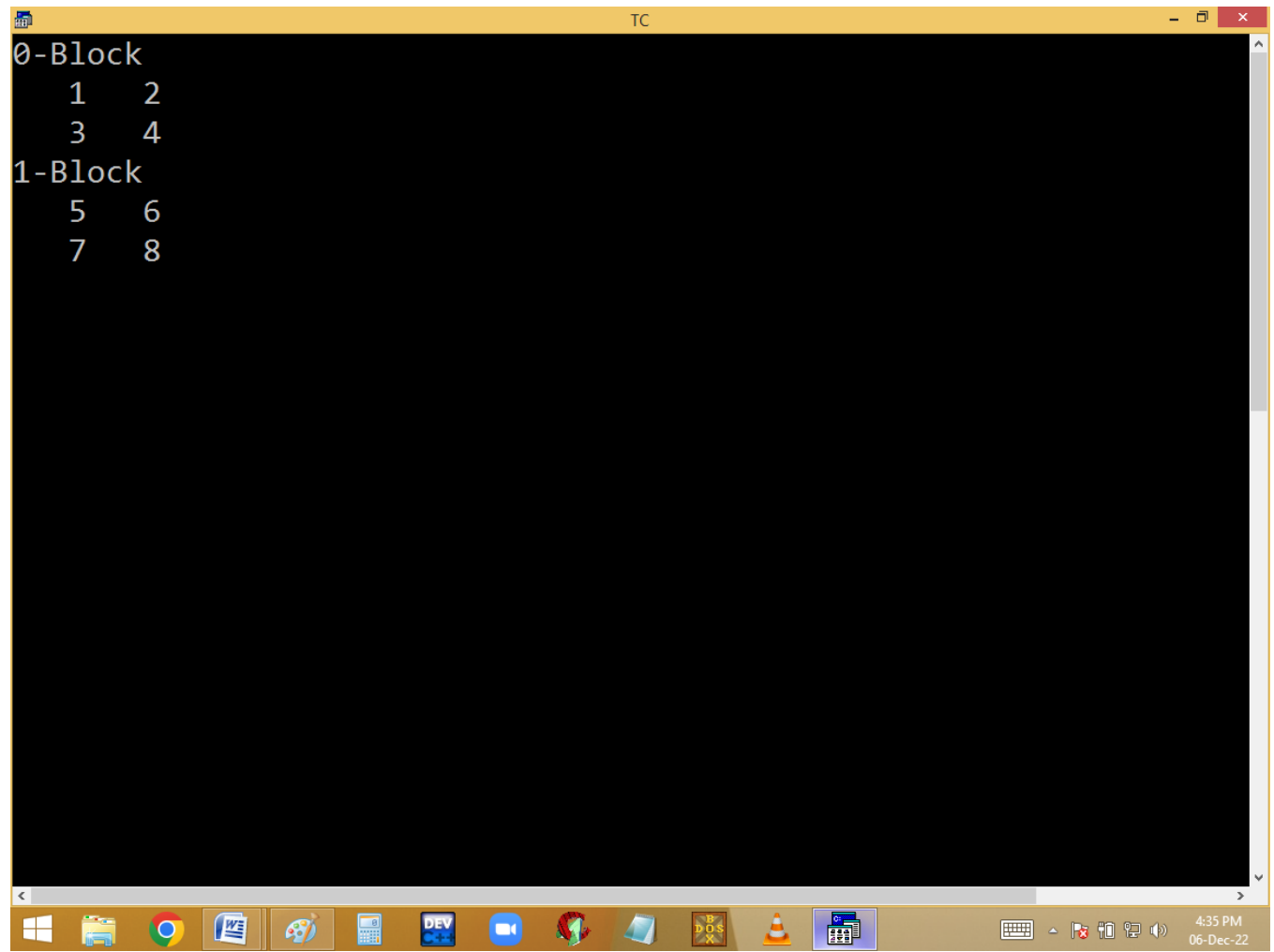
rows
 int a[2][2][2]; an array with 3 subscripts [].
 blocks columns



eg: int class[2][60][6];
 datatype class[sections][stus][marks];

```

TC
File Edit Run Compile Project Options Debug Break/
Edit
Line 1 Col 2 Insert Indent Tab Fill Unindent * E:NONAM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2][2]={1,2,3,4,5,6,7,8},b,r,c;
clrscr();
for(b=0;b<2;b++)
{
printf("%d-Block\n",b);
for(r=0;r<2;r++)
{
for(c=0;c<2;c++) printf("%4d",a[b][r][c]);
printf("\n");
}
}
getch();
}
Watch
  
```



```
TC
0-Block
  1  2
  3  4
1-Block
  5  6
  7  8
```

4-dimensional array:

An array with several sets, blocks, rows and columns.

An array with 4 subscripting operators **[] [] [] []**

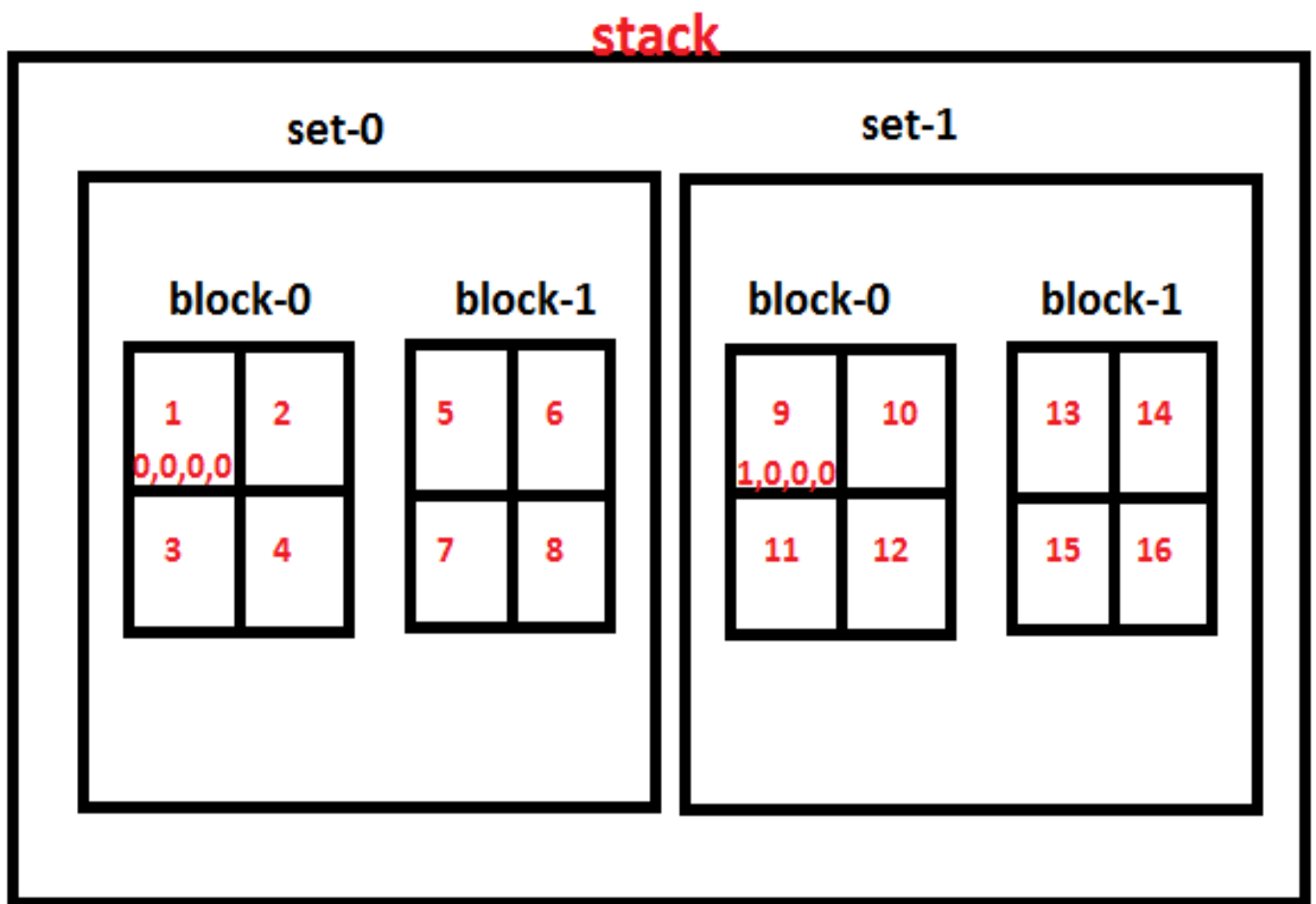
▪

Syntax:

datatype variable [sets] [blocks] [rows] [cols];

eg:

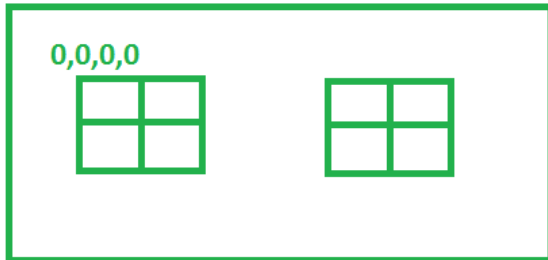
int a[2] [2] [2] [2]= {1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16};



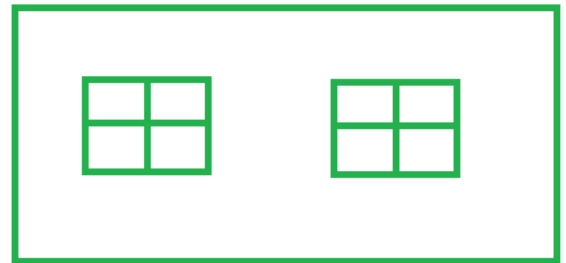
```
datatype var[class][sec][stu][marks];  
int      school[5][2][60][6];
```

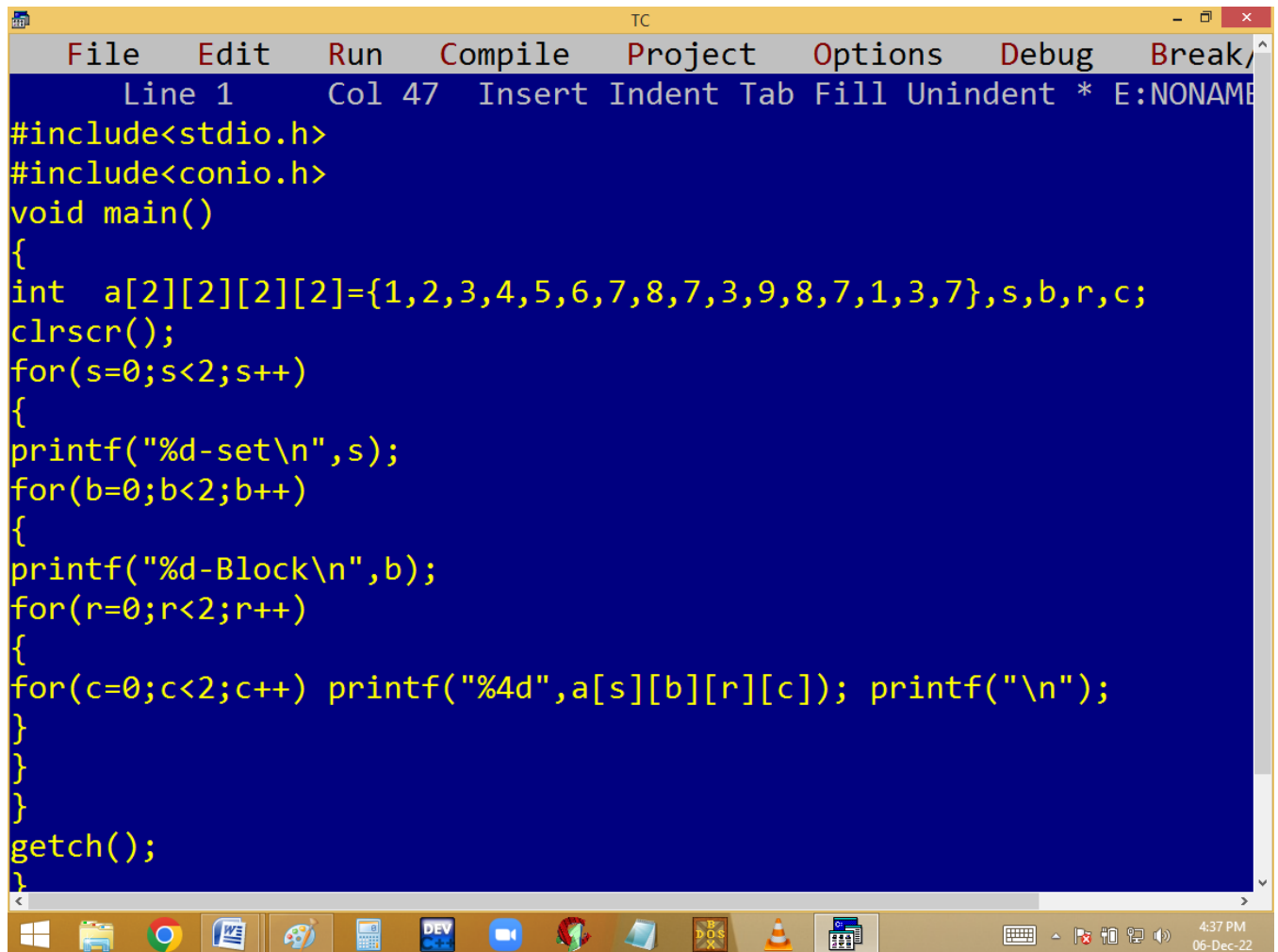
set
rows
int a[2][2][2][2]; an array with 3 subscripts [].
blocks columns

set-0



set-1

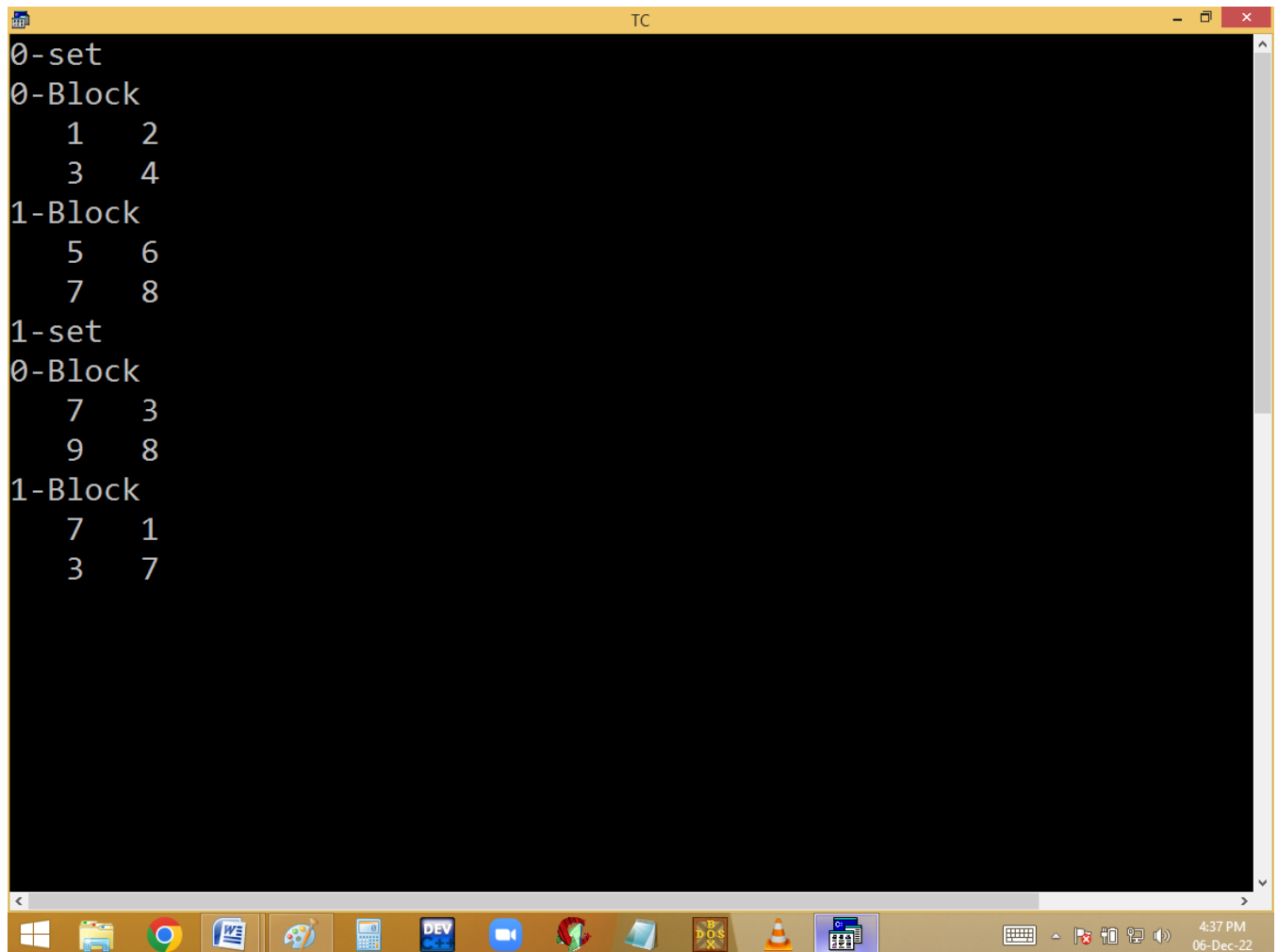




The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/". The status bar at the top indicates "Line 1", "Col 47", and "Insert Indent Tab Fill Unindent * E:NONAME". The main editing area has a dark blue background with yellow text. The code is a C program that defines a 2D array 'a' and prints its elements in a formatted manner. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  a[2][2][2][2]={1,2,3,4,5,6,7,8,7,3,9,8,7,1,3,7},s,b,r,c;
clrscr();
for(s=0;s<2;s++)
{
printf("%d-set\n",s);
for(b=0;b<2;b++)
{
printf("%d-Block\n",b);
for(r=0;r<2;r++)
{
for(c=0;c<2;c++) printf("%4d",a[s][b][r][c]); printf("\n");
}
}
}
getch();
}
```

The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 4:37 PM on 06-Dec-22.



The screenshot shows a Turbo C++ (TC) compiler window with a black background and white text. The text represents assembly code, including instructions like '0-set', '0-Block', '1-Block', and '1-set', along with numerical values. The window has a standard title bar with 'TC' and window control buttons. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 4:37 PM on 06-Dec-22.

```
0-set
0-Block
    1    2
    3    4
1-Block
    5    6
    7    8
1-set
0-Block
    7    3
    9    8
1-Block
    7    1
    3    7
```

STRINGS

A group of characters is called string.

It is alpha-numeric. i.e. we can store both alphabets, numbers and special char's

It is a one dimensional character array.

It is implicit pointer.

It is a derived data type.

Eg:

Char city[]="Hyd-1";

```
int sub[6]={70,88,98,90,69,79};
```

```
char name[50]="baby";
```

```
p("%s",name);
```

